Ventilation Grille for Circular Air Ducts RDR/RDR1



RDR

Description
Rectangular aluminum grille for mounting on circular air ducts.

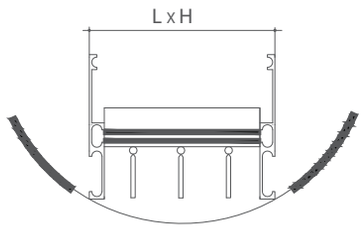
Application
Supply/ exhaust of the air flow.

Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
RDR has two rows of adjustable blades and **RDR1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

Accessories
Air control damper with opposable blades **OBD**.

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request .

Technical design
RDR



Grill Dimension	Ø tube
400 x 100	200
500 x 100	200
600 x 100	250
400 x 150	250
500 x 150	315
600 x 150	400
400 x 200	315
500 x 200	400
600 x 200	400
800 x 200	500
1000 x 200	630

Assembly Method
Standard mounting, with screws.

Selection table RDR/RDR1

Air flow m³/h	LxH	400x100	500x100	600x100	400x150	500x150	600x150	400x200	500x200	600x200	800x200	1000x200
	Ak[m²]	0.013	0.016	0.019	0.025	0.031	0.037	0.033	0.042	0.05	0.067	0.083
350	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	9.3 - 7.48 16.0	8.0 - 6.08 10.0	7.1 - 5.12 7.0	5.9 - 3.89 5.0							
400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	10.7 - 8.55 21.0	9.2 - 6.94 13.0	8.1 - 5.85 9.0	6.7 - 4.44 6.0	5.8 - 3.58 4.0		5.6 - 3.37 3.0				
450	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	12.0 21.0 9.62 27.0	10.3 - 7.81 17.0	9.1 - 6.58 12.0	7.6 - 5.00 8.0	6.5 - 4.03 5.0		6.3 - 3.79 4.0				
500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	13.3 25.0 10.7 33.0	11.5 19.0 8.7 21.0	10.2 - 7.3 14.0	8.4 - 5.6 10.0	7.3 - 4.5 6.0	6.4 - 3.8 4.0	6.9 - 4.2 5.0	6.0 - 3.3 3.0			
600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	16.0 30.0 12.8 47.0	13.8 24.0 10.4 30.0	12.2 19.0 8.8 21.0	10.1 - 6.7 14.0	8.7 - 5.4 9.0	7.7 - 4.5 6.0	8.3 - 5.1 8.0	7.2 - 4.0 5.0	6.4 - 3.3 3.0	5.2 - 2.5 2.0	
700	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	18.7 35.0 15.0 64.0	16.1 29.0 12.2 41.0	14.2 24.0 10.2 28.0	11.8 19.0 7.8 19.0	10.2 - 6.3 12.0	9.0 - 5.3 8.0	9.7 - 5.9 10.0	8.4 - 4.6 7.0	7.4 - 3.9 5.0	6.1 - 2.9 3.0	5.3 - 2.3 2.0
800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	21.3 39.0 17.1 84.0	18.4 33.0 13.9 53.0	16.2 28.0 11.7 37.0	13.5 23.0 8.9 25.0	11.6 - 7.2 16.0	10.3 - 6.0 11.0	11.1 - 6.7 14.0	9.6 - 5.3 9.0	8.5 - 4.4 6.0	7.0 - 3.3 3.0	6.0 - 2.7 2.0
900	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	24.0 42.0 19.2 106.0	20.7 36.0 15.6 67.0	18.3 32.0 13.2 46.0	15.2 21.0 10.0 32.0	13.1 - 8.1 20.0	11.6 - 6.8 14.0	12.5 - 7.6 17.0	10.8 - 6.0 11.0	9.5 - 5.0 8.0	7.9 - 3.7 4.0	6.8 - 3.0 3.0
1000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]		22.9 40.0 17.4 83.0	20.3 35.0 14.6 57.0	16.8 30.0 11.1 40.0	14.5 24.0 9.0 25.0	12.8 19.0 7.5 17.0	13.9 22.0 8.4 21.0	12.0 - 6.6 13.0	10.6 - 5.6 9.0	8.7 - 4.1 5.0	7.5 - 3.3 3.0
1200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			24.4 40.0 17.5 83.0	20.2 35.0 13.3 57.0	17.4 29.0 10.8 36.0	15.4 25.0 9.0 25.0	16.7 27.0 10.1 31.0	14.4 21.0 7.9 19.0	12.7 - 6.7 13.0	10.5 - 5.0 7.0	9.0 - 4.0 5.0
1500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			30.5 47.0 21.9 129.0	25.3 42.0 16.7 89.0	21.8 36.0 13.4 56.0	19.3 31.0 11.3 39.0	20.8 34.0 12.6 48.0	17.9 28.0 9.9 30.0	15.9 23.0 8.3 21.0	13.1 - 6.2 12.0	11.3 - 5.0 7.0
1750	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]					25.4 41.0 15.7 77.0	22.5 36.0 13.1 53.0	24.3 39.0 14.7 65.0	20.9 33.0 11.6 41.0	18.5 28.0 9.7 28.0	15.3 20.0 7.3 16.0	13.2 - 5.9 10.0
2000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]						25.7 40.0 15.0 69.0	27.8 43.0 16.8 85.0	23.9 37.0 13.2 53.0	21.2 32.0 11.1 37.0	17.5 24.0 8.3 20.0	15.0 - 6.7 13.0
2500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]								29.9 44.0 16.5 84.0	26.5 39.0 13.9 58.0	21.8 31.0 10.4 32.0	18.8 25.0 8.4 20.0
3000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									31.8 44.0 16.7 83.0	26.2 37.0 12.4 46.0	22.6 31.0 10.0 29.0
3500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									37.1 49.0 19.4 113.0	30.6 41.0 14.5 63.0	26.0 35.0 11.7 40.0
4000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]											30.0 40.0 13.4 52.0

Vk = 0.25 (m/s) Room height = 2.7 m

NR < 25 | 25≤NR≤35 | NR>35

Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	OBD	/	Dimensions	/	L x H	/	0
RDR / RDR1		Air control damper		L x H 400 x 100, 500 x 100, 600 x 100 400 x 150, 500 x 150, 600 x 150 400 x 200, 500 x 200, 600 x 200, 800 x 200, 1000 x 200		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints